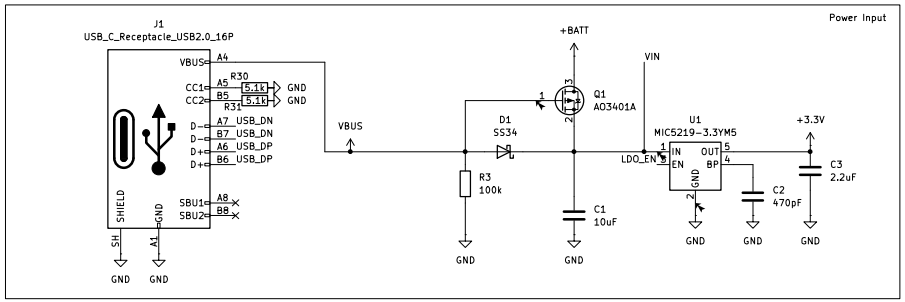
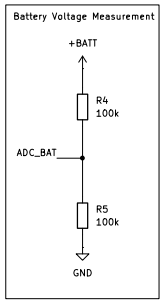
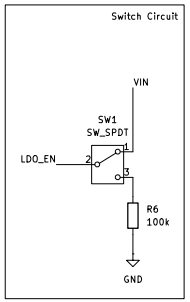
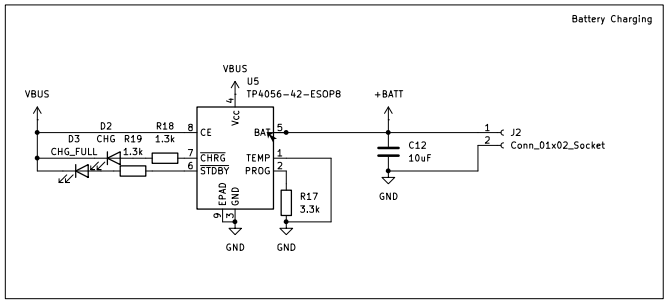
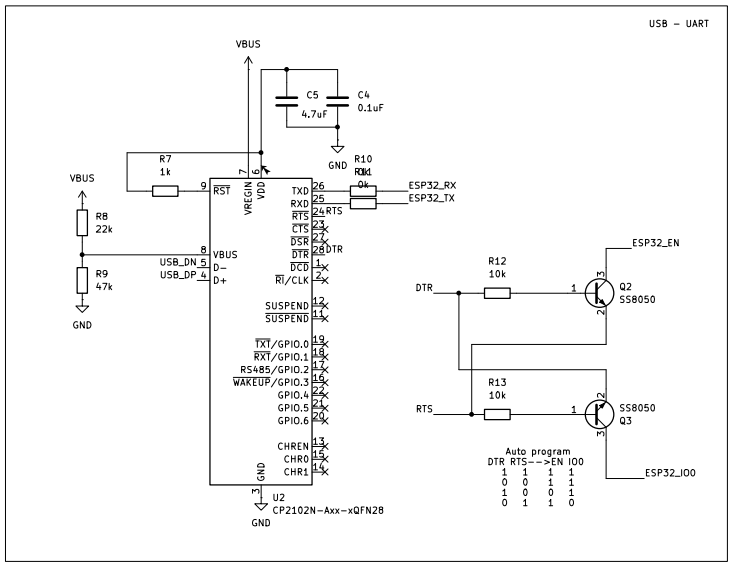
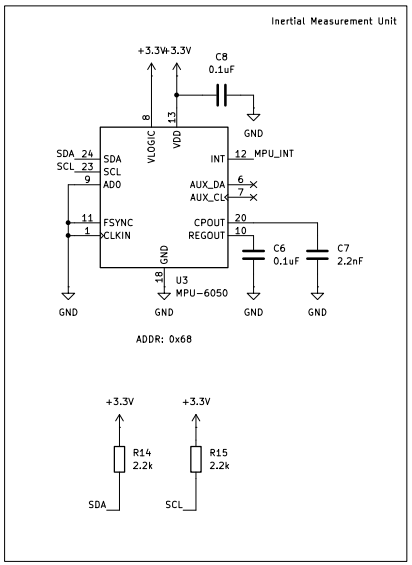

Power Input
J1
USB_C_Receptacle_USB2.0_16P
VBUS
CC1
CC2
R30
5.1k
R31
5.1k
GND
D-
D+
USB_DN
USB_DP
SHIELD
GND
SBU1
SBU2
SH
A1
+BATT
Q1
AO3401A
D1
SS34
R3
100k
C1
10uF
VIN
U1
MIC5219-3.3YM5
IN
OUT
EN
BP
LDO_EN
C2
470pF
C3
2.2uF
+3.3V

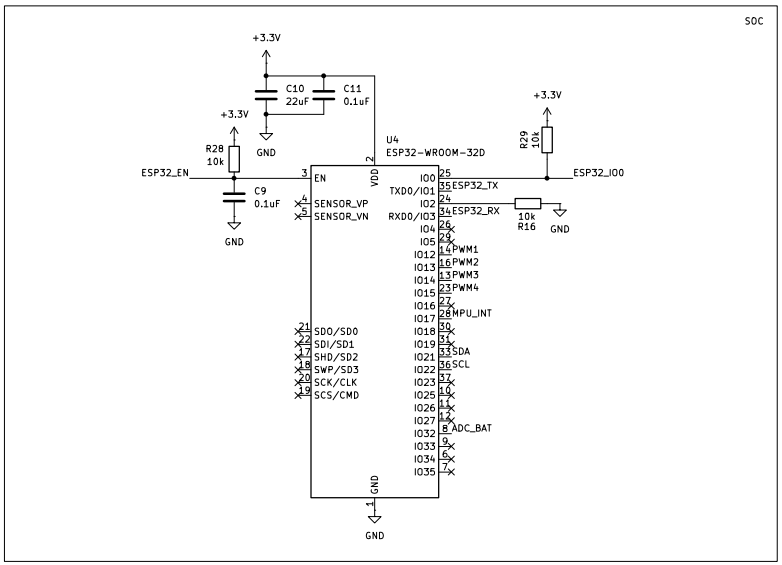

Battery Voltage Measurement
+BATT
R4
100k
ADC_BAT
R5
100k
GND

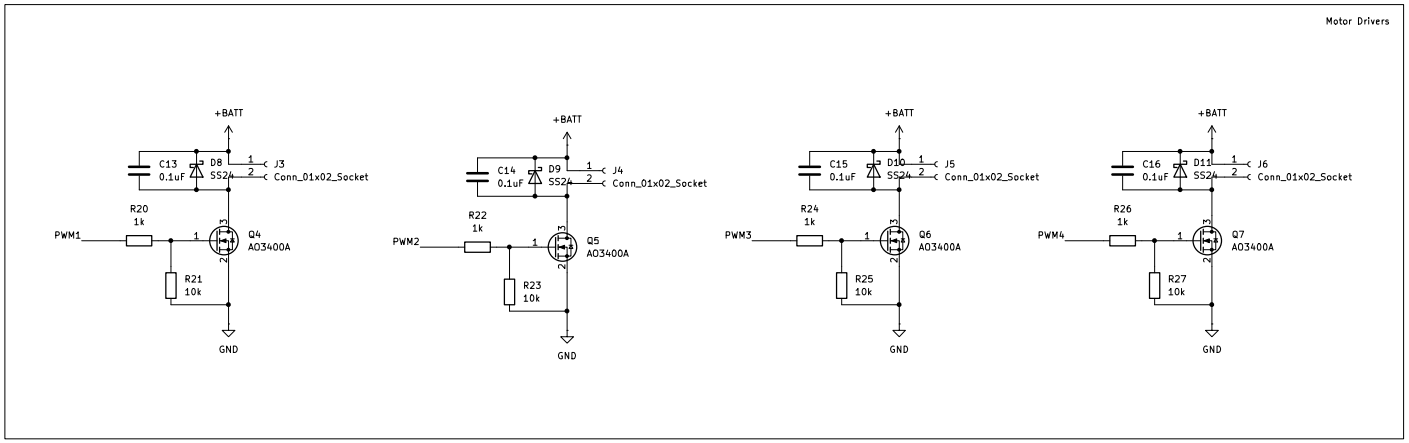

Switch Circuit
VIN
SW1
SW_SPDT
LDO_EN
R6
100k
GND


Battery Charging
VBUS
U5
TP4056-42-ESOP8
+BATT
D2
D3
CHG
CHG_FULL
R19
1.3k
R18
1.3k
CE
BAT
CHRG
STDBY
TEMP
PROG
R17
3.3k
C12
10uF
J2
Conn_01x02_Socket
EPAD
GND


USB - UART
VBUS
C5
4.7uF
C4
0.1uF
R7
1k
R10
0k
R11
0k
ESP32_RX
ESP32_TX
R8
22k
R9
47k
USB_DN
USB_DP
RST
VREGIN
VDD
TXD
RXD
RTS
CTS
DSR
DTR
DCD
RI/CLK
SUSPEND
TXT/GPIO.0
RXT/GPIO.1
RS485/GPIO.2
WAKEUP/GPIO.3
GPIO.4
GPIO.5
GPIO.6
CHREN
CHR0
CHR1
U2
CP2102N-Axx-xQFN28
R12
10k
R13
10k
Q2
SS8050
Q3
SS8050
ESP32_EN
ESP32_IO0
Auto program
DTR RTS-->EN IO0
1 1 1 1
0 0 1 1
1 0 0 1
0 1 1 0


Inertial Measurement Unit
+3.3V
C8
0.1uF
SDA_24
SCL_23
SDA
SCL
AD0
VLOGIC
VDD
INT
MPU_INT
AUX_DA
AUX_CL
FSYNC
CLKIN
CPOUT
REGOUT
C6
0.1uF
C7
2.2nF
U3
MPU-6050
ADDR: 0x68
R14
2.2k
R15
2.2k


SOC
+3.3V
C10
22uF
C11
0.1uF
R28
10k
C9
0.1uF
ESP32_EN
U4
ESP32-WROOM-32D
EN
VDD
SENSOR_VP
SENSOR_VN
IO0
TXD0/IO1
IO2
RXD0/IO3
IO4
IO5
IO12
IO13
IO14
IO15
IO16
IO17
IO18
IO19
IO21
IO22
IO23
IO25
IO26
IO27
IO32
IO33
IO34
IO35
SDO/SD0
SDI/SD1
SHD/SD2
SWP/SD3
SCK/CLK
SCS/CMD
ESP32_IO0
ESP32_TX
ESP32_RX
PWM1
PWM2
PWM3
PWM4
MPU_INT
SDA
SCL
ADC_BAT
R29
10k
R16
10k


Motor Drivers
+BATT
C13
0.1uF
D8
SS24
J3
Conn_01x02_Socket
R20
1k
PWM1
Q4
AO3400A
R21
10k
C14
0.1uF
D9
SS24
J4
Conn_01x02_Socket
R22
1k
PWM2
Q5
AO3400A
R23
10k
C15
0.1uF
D10
SS24
J5
Conn_01x02_Socket
R24
1k
PWM3
Q6
AO3400A
R25
10k
C16
0.1uF
D11
SS24
J6
Conn_01x02_Socket
R26
1k
PWM4
Q7
AO3400A
R27
10k
GND